Progressive exercises

E1 · Practice. In an orthonormal basis of space, consider $\vec{u} = (2, -1, 3)$ and $\vec{v} = (1, 4, 0)$. Calculate $\vec{u} \cdot \vec{v}$.

E2 · Practice. In an orthonormal basis, consider $\vec{u} = (a, 2, 1)$ and $\vec{v} = (3, -1, 4)$, where a is real. Find a so that $\vec{u}$ and $\vec{v}$ are orthogonal.

E3 · Practice. In an orthonormal frame, a plane is spanned by $\vec{u} = (1, -1, 0)$ and $\vec{v} = (1, 0, -1)$. Show that $\vec{n} = (1, 1, 1)$ is normal to it. Then consider $A(0, 0, 0)$ and $B(2, 2, 2)$. Describe geometrically the set of points equidistant from A and B.

E4 · Practice. In an orthonormal basis, consider $\vec{u} = (1, 0, 1)$ and $\vec{v} = (0, 1, 1)$. Find the measure, between 0 and π, of the angle between these two vectors.

E5 · Practice. A tetrahedron has as its base a right-angled triangle whose two perpendicular sides measure 3 cm and 4 cm. The height of the tetrahedron relative to this base is 5 cm. Calculate its volume in cm^3.

Textbook or past-paper work

Barbazo, p. 362, exercises 23 and 27. In S48: exercise 23, to calculate a dot product in coordinates. At home after S54: exercise 27; check orthogonality to the two direction vectors of a plane, without settling for a single zero product.